

EXAM 7 – SPRING 2011

17. (3 points)

An insurance company that writes catastrophe coverage is considering the use of reinsurance in 2012 in order to reduce risk.

Expected premium, loss and expense for 2012 are as follows:

- Expected gross premium: \$500 million
- Expected gross losses: \$150 million
- Expected gross expenses: \$60 million

Three possible reinsurance options are being considered as follows:

	<u>Option 1</u>	<u>Option 2</u>	<u>Option 3</u>
Expected reinsurance premium	No	\$250 million	\$100 million
Expected reinsurance recoveries	Reinsurance	\$80 million	\$30 million

Assume no reinsurance commission expense and the cost of capital is 12%.

The distribution of 2012 modeled underwriting profit is partially described as follows (\$000,000):

<u>Percentile</u>	<u>Gross</u>	<u>Option 2</u>	<u>Option 3</u>
90.00%	250	(15)	100
95.00%	(10)	(50)	(200)
99.00%	(1,400)	(300)	(400)
99.50%	(2,200)	(420)	(600)
99.90%	(4,500)	(2,000)	(2,500)

a. (0.75 point)

Use an expected value basis to determine which option the company should select.

b. (2.25 points)

The company requires capital equal to 1.5 times the 1-in-200VaR. Based on the allocated risk capital methodology, determine the savings in capital cost of both options 2 and 3.

CONTINUED ON NEXT PAGE

Sample Solutions

Exam 07, 2011, Question 17

Solution 1

a)

Gross - Option 1: UW Profit = $500 - 150 - 60 = 290$

Option 2: UW Profit = $500 - 150 - 60 - 250 + 80 = 120$

Option 3: UW Profit = $500 - 150 - 60 - 100 + 30 = 220$

Since the gross option results in the highest UW profit, option 1 should be selected by the company

b)

	Gross	Option 2	Option 3
Risk Capital Estimate	$1.5 \times 2,200 = 3,300$	$1.5 \times 420 = 630$	$1.5 \times 600 = 900$
Capital Cost	$.12 \times 3,300 = 396$	$.12 \times 630 = 75.6$	$.12 \times 900 = 108$
Savings in Capital Cost		$396 - 75.6 = \mathbf{320.4}$	$396 - 108 = \mathbf{288}$